

供者来源游离 DNA 与多组学标志物在实体器官移植排斥监测中的应用价值与未来方向

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Abstract

供者来源游离 DNA (donor-derived cell-free DNA, dd-cfDNA) 已经成为近五年实体器官移植无创监测中最接近临床常规化的分子工具之一。其优势是对移植物细胞损伤高度敏感、可纵向重复采样, 并能在肾、心、肺、肝等器官中与排斥、微血管炎症、供者特异性抗体和病理损伤形成关联。其弱点同样清楚: dd-cfDNA 反映“损伤释放”, 并不天然区分抗体介导排斥、T 细胞介导排斥、缺血再灌注损伤、感染、药物毒性或机械性损伤。未来 3-5 年, 最可行且新颖的路线不是把 dd-cfDNA 作为活检替代品, 而是把它与 DSA、尿液外泌体 mRNA、转录组、蛋白组、病毒负荷、免疫抑制暴露和机器灌注标志物整合成动态风险模型, 用于决定何时活检、何时减免疫、何时强化治疗以及何时避免过度诊断。

1 引言

传统移植监测依赖血清肌酐、肝酶、心功能、肺功能、DSA 和组织活检。问题在于功能指标滞后且不特异, 活检有侵入性、采样误差和读片差异, DSA 能提示免疫风险但不能直接量化当前组织损伤。dd-cfDNA 的临床吸引力来自它把移植物细胞死亡转化为外周血可测信号。ADMIRAL 等纵向研究推动了肾移植 dd-cfDNA 从研究工具走向临床监测; Trifecta 研究进一步把 dd-cfDNA 与分子病理联系起来, 提示它更像“损伤强度计”而非单一排斥标签 [1, 2]。

近年证据也显示, dd-cfDNA 在心、肺、肝和多器官移植中迅速扩展, 但不同器官的基线、释放动力学、取样频率和干扰因素差异明显 [3-5]。因此, dd-cfDNA 的临床价值必须在器官特异性场景中解释, 而不是用同一阈值横跨所有移植类型。

2 肾移植: 证据最丰富, 也最能暴露边界

肾移植是 dd-cfDNA 临床验证最充分的领域。ADMIRAL 研究显示, dd-cfDNA 纵向升高可与排斥和不良结局相关; 一些研究支持把 dd-cfDNA 与肌酐、蛋白尿、DSA 和临床事件联合解读, 以提高对早期损伤的敏感性 [1, 6]。在抗体介导排斥 (ABMR) 中, dd-cfDNA 往往比单纯功能指标更接近微血管损伤过程, 尤其当 DSA 或 AT1R 抗体提示免疫风险时更有解释力 [7, 8]。

但肾移植也提醒我们不能过度承诺。dd-cfDNA 升高可见于感染、BK 病毒相关损伤、缺血再灌注、钙调神经磷酸酶抑制剂毒性、复发性肾病和近期手术损伤。分数阈值在不同平台、不同人群和不同时间点之间并非完全可移植。分数比例还可能受受者总 cfDNA 背景影响; 部分研究因此同时讨论 fraction 与 absolute quantity 的价值 [9]。未来模型若只输出“阳性/阴性”, 很难服务复杂门诊决策; 更合理的是输出时间趋势、个体基线偏离和与其他证据的一致性。

3 心、肺、肝与多器官移植: 从可测到可解释

心移植长期依赖心内膜活检, 但活检频繁、侵入性强且对局灶病变敏感性有限。dd-cfDNA 在心移植中可与急性排斥、移植物损伤和 DSA/ABMR 风险形成关联; 近期研究也开始比较 dd-cfDNA 与心肌活检基因表达、线粒体 DNA 和人群差异之间的关系 [10-12]。关键挑战是心衰、机械循环支持、感染和缺血损伤也会改变 cfDNA 背景。

肺移植中，感染、原发移植物功能障碍和慢性肺移植物功能障碍会让 dd-cfDNA 解释更复杂，但其连续监测价值很高，因为肺移植临床事件密集且组织采样负担重 [13]。肝移植中，肝细胞体量大、再生能力强、手术早期释放高，dd-cfDNA 的诊断阈值和时间窗需要单独建模；系统综述和荟萃分析正在推动这一领域从探索性研究进入诊断准确性评价 [14, 15]。多器官移植则进一步提出来源归属问题：当心肾、肝肾或胰肾同时移植时，dd-cfDNA 的器官来源、比例解释和干预决策必须依赖更精细的供体基因型和器官特异标志物。

4 多组学整合：让 dd-cfDNA 从“损伤读数”变成“病因推断”

dd-cfDNA 最大的临床短板是病因特异性不足，多组学的意义正是在这里。尿液外泌体 mRNA、尿液趋化因子、外周血转录组、蛋白组、代谢组、DSA、HLA eplet mismatch 和病理分子分类，可以为 dd-cfDNA 提供免疫通路解释 [16–18]。例如，dd-cfDNA 升高并伴随 DSA、微血管炎症和 ABMR 分子特征时，临床处理应不同于 dd-cfDNA 升高但感染证据充分的情境。

机器灌注标志物可成为移植前层面的补充。扩展标准供体器官、DCD 器官或高缺血风险移植物在植入前已经释放代谢和炎症信号；若这些灌注液指标与植入后 dd-cfDNA 动态连接，就可能区分“可预期的保存损伤”与“异常免疫损伤”。这对减少不必要活检和避免过早治疗尤其重要。

5 讨论：临床应用场景与解释边界

第一个场景是低风险稳定患者的活检减量。若 dd-cfDNA 长期低位、DSA 阴性、功能稳定且尿液/转录组无炎症信号，协议活检的收益可能下降。第二个场景是高风险患者的早期预警。免疫不依从、DSA 新发、既往 ABMR、边缘供体器官或感染后免疫调节患者，dd-cfDNA 趋势可帮助决定活检时点。第三个场景是治疗反应监测。ABMR 或 TCMR 治疗后，dd-cfDNA 是否下降可能比单一功能指标更快反映组织损伤缓解 [19, 20]。第四个场景是免疫抑制精准化。dd-cfDNA 与病毒负荷、TTV、药物浓度和感染风险联合，可服务“减免疫”与“防排斥”的平衡。

6 展望：未来 3-5 年研究方向

1. 建立器官特异、时间特异的动态阈值。早期术后、稳定期、感染期、DSA 新发期和治疗后不应使用同一解释框架。
2. 同时报告 fraction、absolute quantity 和个体化变化率。比例值容易受受者 cfDNA 背景影响，绝对量与趋势可提供互补信息。
3. 开展 dd-cfDNA + DSA + 尿液外泌体/趋化因子 + 分子病理的前瞻性决策试验。终点不应只看诊断准确性，还应看活检减少、误治减少、排斥延误减少和长期功能。
4. 把机器灌注标志物纳入早期损伤解释。对 DCD、扩展标准供体和高冷缺血器官，术前灌注液读数可作为 dd-cfDNA 术后曲线的先验。
5. 发展可解释 AI 风险模型。模型应输出“可能病因组合”和“建议下一步证据”，而不是黑箱风险分。临床可用性取决于它能否告诉医生该活检、复查、调整免疫抑制还是寻找感染。

7 结论

dd-cfDNA 已经改变移植随访的语言：医生不再只能等待器官功能变差，而能更早看到移植物细胞损伤的分子信号。但 dd-cfDNA 不是活检的简单替代品，也不是排斥的单一同义词。它最有前景的位置，是作为多证据动态监测系统中的核心损伤轴，与免疫风险、感染风险、组织/尿液组学和移植前器官质量共同构成临床决策模型。未来 3-5 年，真正能进入常规路径的方案，将是那些能减少不必要活检、避免过度治疗并改善长期结局的整合型监测策略。

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